

Wordglass Build Plan

18 slices · design → build → review

A step-by-step sequence for building V1. Every slice runs the same loop: **Claude Code scopes it, emits design prompts, screens get designed, Claude Code builds against them, you review.**

SLICES	LOOP	TOTAL
18 Two foundation, sixteen feature	6 steps Scope → prompt → design → build → sweep → commit	22–27 weeks Solo build
DESIGN OUTPUT	GATE	METHOD
Screens + states Never screens alone	Fidelity sweep Every UI slice, no exceptions	Spec-driven specify → plan → tasks → implement

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The Loop

Six steps. Identical on every slice. The value is in the repetition — variation is where quality leaks out.

1

Scope the slice

Claude Code runs `/specify`. Output: goal, user-visible behaviour, acceptance criteria, explicit out-of-scope list. No screens named yet — behaviour first, or the design gets anchored on a guess.

2

Emit design prompts

Claude Code enumerates every screen *and every state of every screen*, then writes one design prompt per screen. Template in Section 02. This is the step most likely to be rushed; a thin prompt produces a screen that has to be redesigned after build.

3

Design the screens

Screens produced from the prompts, checked against the state matrix in Section 03 and the principles in Section 04. Output: visual for every state, plus any new tokens or components that did not already exist.

4

Write the handoff

Designs converted into something Claude Code can build deterministically: token names not hex values, component names not descriptions, exact copy strings, state transition table, measurements in dp. A screenshot alone is not a handoff.

5

Build

Claude Code runs `/plan` → `/tasks` → `/implement`, uncommitted. Handoff document is the source of truth, not the visual.

6

Fidelity sweep, then commit

Section 05. Screenshot each built state against its design. Fix, then commit. Nothing commits without passing the sweep.

WHY STEP 4 EXISTS

Handing a visual straight to the builder is the single most common failure in this workflow. The model infers values that were never specified — spacing, exact copy, disabled-state colour, what happens on long-press — and each inference is a silent decision you did not make. The handoff document turns inference into lookup. It takes fifteen minutes and removes most of the rework.

1.1 What not to design

Time spent designing these is wasted — they are system-owned and cannot be restyled:

- The Android runtime permission dialog. You design the *pre-prompt* that precedes it and the *denied* state that follows, never the dialog itself
- The system share sheet
- The Storage Access Framework file picker

- The Google Play purchase flow
- System notifications beyond icon, title, body, and actions

Design Prompt Template

What Claude Code should produce in step 2. One of these per screen, not per slice.

```
# Screen: [name]           Slice: [n]           Platform: Android / Compose
```

PURPOSE

One sentence. What the user is trying to accomplish here.

ENTRY / EXIT

Arrives from: [screen or event]

Leaves to: [screens, and what triggers each]

Back behaviour: [what the system back gesture does – including
mid-recording and mid-export]

STATES TO DESIGN

[list every applicable state from the Section 03 matrix,
with a one-line description of each]

CONTENT INVENTORY

Every string, verbatim. Every number and its format.

Longest realistic value for each field (for overflow).

Empty-state copy and its call to action.

CONTROLS

Each control: label, icon, enabled/disabled conditions,
destructive yes/no, confirmation required yes/no.

CONSTRAINTS

Thumb-reach: which controls must sit in the bottom third

One-handed: yes/no

Glanceable: is the user looking at the screen or the lens

Over live camera: yes/no (drives contrast requirements)

Longest-running operation and whether it can be backgrounded

REUSE

Existing components this must use: [names from the design system]

New components this justifies: [name + why nothing existing fits]

OUT OF SCOPE

Explicitly not on this screen.

THE TWO FIELDS THAT CARRY THE MOST WEIGHT

Longest realistic value and **Reuse**. The first prevents the layout that looks perfect with "My Script" and breaks with a real 74-character title. The second is what stops slice 9 from inventing a fourth kind of slider. Without an explicit reuse field, every screen quietly grows its own component set and the app stops looking like one product somewhere around slice 6.

Universal Screen States

Design produces states, not screens. A screen designed only in its happy state is a third of a screen.

STATE	APPLIES WHEN
Empty	No content yet. Needs copy and a call to action, not a shrug
Loading	Local reads are fast enough to skip. Anything network-bound needs a designed state
Populated	Design at realistic volume — 40 scripts, not 3
Overflowing	Longest realistic string in every field. Long titles, long transcripts, long org names
Error — retryable	Network failed. Must offer the retry, not just report the failure
Error — terminal	DRM-protected audio, unsupported codec, quota exhausted. Explain in plain language and offer the alternative
Offline	Recording works offline. Captions queue. The UI must say which
Permission pre-prompt	Before the system dialog. Explains why, in the user's terms
Permission denied	Soft denial — ask again with context
Permission denied permanently	Deep link to Settings. Different screen, different copy
Working / progress	Anything over ~2s. Honest progress, cancellable where possible
Quota reached	Free-tier caption minutes or generations exhausted. This is a paywall moment — design it deliberately
Interrupted	Call arrived, app backgrounded, storage full. What the user sees on return

Not every state applies to every screen. The prompt should list which do and say nothing about the rest — an unmentioned state is a state nobody built.

UX Principles for This App

Six constraints specific to Wordglass. Where these conflict with a generic mobile pattern, these win.

4.1 The user is not looking at the screen

During recording, the user's eyes are on the lens. This is the single most consequential fact about the interface and it changes almost everything in the capture UI.

- Prompter text must be readable at arm's length in peripheral vision — larger than any type scale you would otherwise pick, and adjustable
- Controls must be findable without looking. That means fixed positions, generous hit targets well above the 48dp minimum, and haptic confirmation instead of visual
- Nothing may move position between states. A button that shifts when recording starts is a button the user cannot hit
- Recording status must be perceivable in peripheral vision — a persistent edge treatment, not a small dot

4.2 Dark-first, and camera-aware

A light UI beside a live camera preview blows out the user's perception of the exposure they are actually recording. The app is dark-themed throughout — not a dark *mode*, a dark product. Chrome over the preview must hold contrast against an unpredictable background, which means scrim and outlines rather than colour alone. Design every over-camera element twice: against a white wall and against a dark room.

4.3 Takes are precious

A user who has just delivered a good 90-second read will not cheerfully do it again. Every destructive action on a take needs confirmation or undo. Auto-save on interruption is a feature, not error handling. Never let a network failure, a filter change, or a navigation mistake destroy a recording.

4.4 Local first, always

Recording, trimming, take management, and playback work with no connectivity. Only captions, generation, and sync require the network. The UI should never block a local action on a remote call, and should never present a spinner for something that is actually instant.

4.5 Honest progress

A ten-minute 1080p export on a mid-range device takes minutes, not seconds. Indeterminate spinners are dishonest here. Show real percentage, show elapsed and estimated remaining, allow backgrounding, notify on completion. This is the difference between a slow app and an app that "freezes" in the reviews.

4.6 Two personas, one app

The **writer** works on the web, in bulk, at a desk. The **recorder** works on mobile, one script at a time, often in a hurry. A solo user is both. Do not force the recorder through writer-oriented UI: the assignment inbox should get someone from notification to recording in two taps.

DESIGN SYSTEM DECISIONS TO MAKE ONCE, IN SLICE 0.5

Type scale with an explicit prompter tier well above body text. Colour tokens split into *surface* and *over-camera* sets. Spacing scale. One slider component that serves beauty, audio mixing, and prompter speed. One bottom-sheet pattern. One progress pattern. One destructive-confirm pattern. Decide these before slice 1 and every later slice becomes assembly rather than invention.

SECTION 05

Fidelity Sweep

The gate between build and commit. Runs on every slice that produces UI. No exceptions, including slices that "only changed one thing".

1

Screenshot every designed state on device

Not the happy path only. If the design has nine states, the sweep has nine screenshots.

2

Compare against the handoff, not the visual

Check the specified values: token names, dp measurements, exact copy strings. Colour picked from a screenshot is not evidence.

3

Check for invented content

Any string, label, icon, or affordance present in the build but absent from the handoff. This is the most common defect and it is always silent.

4

Check for missing states

Any designed state with no code path to reach it.

5

Check reuse

Did the build create a new component where an existing one was specified? Revert it.

6

Over-camera contrast pass

Capture-UI slices only. Screenshot against a white wall and a dark room.

7

Fix, re-sweep, commit

Fixes go in the same uncommitted working set. Commit only after a clean pass.

WHY THIS GATE EXISTS

First-pass builds invent content. Placeholder copy that reads plausibly, an extra menu item that seems helpful, a label reworded to sound better. Each one is individually defensible and collectively they produce an app nobody designed. The sweep is cheap; discovering it at slice 12 is not.

Slice Sequence

Eighteen slices in dependency order. Each is independently reviewable and, from slice 4 onward, independently demonstrable.

SLICE 00 Project foundations

1 wk · no UI

Everything that is painful to retrofit. No screens, no design step.

DESIGN

- None — skip steps 2 to 4 of the loop

BUILD

- Compose project, `minSdk 26`, `targetSdk 36`
- Hilt, Room, Coroutines, Navigation
- `CLAUDE.md` constitution with the fidelity-sweep rule written in
- Supabase project; first migration with `org_id` on every table and RLS policies
- Sentry, PostHog wired but silent
- CI: build + lint on push
- Debug `applicationIdSuffix = ".dev"`

DONE WHEN

A debug build installs alongside nothing, CI is green, and a manual insert into Supabase is blocked by RLS for the wrong `org_id`.

Decide once what every later slice assembles from. This slice is design-heavy and build-light.

DESIGN

- Colour tokens: surface set and over-camera set
- Type scale including a dedicated prompter tier
- Spacing, radius, elevation scales
- Components: button, icon button, slider, bottom sheet, list row, chip, dialog, toast, progress, empty state
- Motion: durations and easing, two values not ten
- Icon set selection

BUILD

- Compose `Theme` with the token set
- Each component as a composable with a `@Preview` per state
- A gallery screen listing every component — debug builds only

DONE WHEN

The gallery screen renders every component in every state, and no slice-1 screen needs a colour or size that is not already a token.

SLICE 01 **Scripts — list and editor**

The easiest real feature. Its purpose is to prove the design-to-build loop works before the loop meets the camera.

DESIGN

- Script list: empty, populated (40 items), overflowing title
- Script editor: new, editing, unsaved-changes
- Delete confirmation
- Word count and estimated read time treatment

BUILD

- Room `Script` entity, DAO, repository
- List and editor screens, navigation
- Autosave on pause
- Read-time estimate at 140 wpm

DONE WHEN

You can write a script, kill the app, reopen it, and the script is there with a correct read-time estimate.

SLICE 02 Teleprompter capture

2.5 wks

The core interaction and the highest-risk slice in the project. Do not compress it.

DESIGN

- Camera permission pre-prompt; denied; denied-permanently
- Capture HUD: idle, countdown, recording, paused
- Prompter overlay: speed, size, opacity, scrim
- Settings sheet: ratio, resolution, mirror, countdown
- Exposure and focus affordances
- Audio level meter
- Interrupted-recording recovery
- **Every over-camera element against white wall and dark room**

BUILD

- CameraX preview, VideoCapture, aspect-ratio guide frame
- Scrolling overlay as a View above the preview — not composited into the video
- Tap-to-pause, drag-to-reposition, mirror mode
- Front/rear switch mid-session, exposure lock, tap-to-focus
- Level meter, headphone monitoring
- Keep-screen-awake; auto-save on interruption
- Room **Take** entity linked to script

DONE WHEN

You record a 90-second read outdoors, take a phone call mid-recording, return, and the partial take is intact and playable.

RISK

Device fragmentation in CameraX behaviour. Test on both target devices before declaring the slice done, not at hardening. Budget two extra days for device-specific quirks.

SLICE 03 Takes — review and manage

0.5 wk

Somewhere to put what slice 2 produces.

DESIGN

- Take list per script: empty, populated, with durations and thumbnails
- Take player with scrub
- Rename, delete with undo
- Storage-used indicator

BUILD

- ExoPlayer playback surface
- Thumbnail extraction and caching
- Delete with undo window, file cleanup on commit

DONE WHEN

Three takes of one script are listed, playable, and a deletion can be undone within the window.

SLICE 04 Export pipeline

1.5 wks

First end-to-end value: script in, video file out. Everything downstream hangs off this pipeline, so it comes before captions and effects.

DESIGN

- Export sheet: quality, aspect ratio, multi-ratio toggle
- Progress: determinate percentage, elapsed, estimated remaining, cancel
- Backgrounded notification
- Complete: preview, share, saved-to-gallery confirmation
- Failure: out of storage, encoder failure

BUILD

- Media3 `Transformer`, `Composition`, `EditedMediaItemSequence`
- WorkManager + foreground service
- MediaStore write
- System share sheet via `ACTION_SEND` + FileProvider
- Multi-ratio export in one pass

DONE WHEN

A 5-minute take exports at 1080p while the app is backgrounded, notifies on completion, and appears in the device gallery.

RISK

This is where you first learn your real export times. Measure on the mid-range device and, if they are worse than expected, revisit the quality defaults before building effects on top.

SLICE 05 Trim and timeline

1 wk

The editing surface that captions and smart clipping will both extend.

DESIGN

- Timeline with waveform: idle, scrubbing, handles engaged
- Trim handles and their touch targets
- Split and delete interior section
- Undo / redo affordance
- Multi-take sequence reorder

BUILD

- Waveform extraction and caching
- Frame-accurate scrub via `CompositionPlayer`
- Trim and split mapped to `EditedMediaItem` clipping
- Undo stack

DONE WHEN

Head and tail trim plus one interior cut survive an export, and undo returns to the previous edit exactly.

SLICE 06 Captions — generation and correction

2 wks

First network-dependent feature. Introduces async state, quota, and the transcript editor that slice 10 depends on entirely.

DESIGN

- Generate entry point and its cost/quota disclosure
- Queued, uploading, processing, complete, failed
- Offline: queued-until-connected
- Transcript editor: word-level selection, correction, timing nudge
- Caption overlay preview on video
- Quota exhausted — first paywall moment

BUILD

- Signed upload to R2 via Edge Function
- AssemblyAI submission and callback webhook
- Word-level timestamp storage in Room and Postgres
- Transcript editor with re-sync on edit
- Caption rendering: Canvas → Bitmap → `BitmapOverlay`
- Usage metering against the entitlement table

DONE WHEN

A take with two deliberately mispronounced proper nouns produces captions you can correct, and the corrections appear burned into the export.

SLICE 07 Caption styling and brand kit

1 wk

Turns captions from functional into branded. Establishes the brand kit that AI generation later consumes.

DESIGN

- Five caption themes, each shown on video
- Theme picker
- Brand colour picker
- Logo upload, position, size
- Per-caption manual repositioning
- Captions off

BUILD

- Theme definitions as data, not hardcoded layouts
- Highlight styles: karaoke, pop-on, single-line
- Logo overlay compositing
- Brand kit persisted at org level

DONE WHEN

Switching theme and brand colour changes the export, and the logo lands where it was placed at all three aspect ratios.

SLICE 08 Beauty and camera effects

2.5 wks

Off by default, user-adjusted only. The shader is written once and used in both preview and export.

DESIGN

- Effects sheet: master toggle off, three sliders at zero
- LUT gallery with *None* selected
- Before/after compare affordance
- Reset to default
- Applied-at-export indicator on existing takes
- Unsupported-device fallback

BUILD

- GLSL: skin smooth, warmth, brightness lift
- ML Kit face contours (thin variant) for the mask
- `CameraEffect` + `SurfaceProcessor` for preview
- Same source as Media3 `GLEffect` at export
- `SingleColorLut` for grade presets
- Per-frame timing instrumentation

DONE WHEN

Preview holds under 30ms per frame at 30fps on the mid-range device with all three sliders at maximum, and the exported file matches the preview.

RISK

The 30ms budget is the hard constraint. Validate it with a trivial shader before writing the real one. If a mid-range device cannot hold it with the mask enabled, reduce mask resolution before reducing the feature.

SLICE 09 Audio import and mixing

0.5 wk

Small slice, one notable failure case that must be designed rather than caught.

DESIGN

- Import entry point
- Mixer: voice and bed levels, ducking toggle
- Bed trim, fade in/out
- Mute original audio
- **DRM-protected file — plain-language terminal error**

BUILD

- SAF picker, `audio/*`
- `AudioProcessor` mixing
- Auto-duck -12 to -15dB using caption timestamps to locate speech
- Explicit DRM/unreadable-source handling

DONE WHEN

A bed ducks correctly under speech, and attempting to import from Spotify produces a clear explanation rather than a crash.

SLICE 10 Smart clipping

2.5 wks

The differentiator. Entirely dependent on slice 6 being solid — do not start it early.

DESIGN

- Filler review: strike-through in transcript, remove-all vs individual
- Silence threshold control with live cut count
- Segment cards: ranked, with hook preview and duration
- Target length selector
- Auto-reframe preview with manual crop override
- **Every automated cut visible and reversible before export**

BUILD

- Filler detection from word timestamps
- Silence detection, default 700ms threshold
- LLM segment extraction via Edge Function, cut on word boundaries
- Face tracking + low-pass-filtered pan → Media3 **Crop**
- Cut list as a reversible edit layer, not a destructive operation

DONE WHEN

A 3-minute take yields three ranked clips, every automated cut is individually reversible, and the reframe pan does not visibly jitter.

SLICE 11 Auth and organisations

1.5 wks

Turns on the UI for a schema that has been org-scoped since slice 0.

DESIGN

- Sign in / sign up, email + Google
- Email verification
- Org creation and naming
- Profile and account settings
- Sync status indicator
- Sign-out with local-data warning

BUILD

- Supabase Auth
- Org created on signup — solo user is an org of one
- Script and metadata sync; video stays local
- Conflict resolution: last-write-wins with local precedence

DONE WHEN

Scripts written offline on one device appear on a second device after sign-in, and RLS blocks cross-org reads.

SLICE 12 AI script generation

1.5 wks

Solves the blank page. Sits above the teleprompter as the real entry point to the product.

DESIGN

- Generation input: topic, duration, tone, audience
- Generating state
- Variant picker — two or three, side by side
- Paragraph-level regenerate
- Inline rewrite on selection
- Import: paste, URL, file, Google Docs
- URL → "turn this into a 60-second script" step
- Quota exhausted — second paywall moment

BUILD

- Edge Function calling the Anthropic API
- Brand kit injected as context
- Word count targeted to duration, reconciled with prompter scroll speed
- SAF import for `.txt`, `.docx`
- Generation metering

DONE WHEN

A generated 30-second script reads aloud naturally at the estimated duration and contains none of the banned corporate filler.

RISK

Output quality is the whole feature. Test by reading three generations aloud on camera before shipping. If it sounds like an LLM, the constraint prompt is wrong and no amount of UI fixes it.

SLICE 13 Free tier, watermark, metering

1 wk

Makes the business model real before billing exists. Metering first, payment later.

DESIGN

- Usage meter: caption minutes and generations remaining
- Approaching-limit state
- Paywall — one design, reached from both quota triggers
- Watermark preview before export
- Watermark-free export as the paid difference

BUILD

- Provider-agnostic `subscription` and `usage` tables
- Watermark as a `BitmapOverlay`, positioned to survive 9:16 platform crops
- Server-side quota enforcement, not client-side
- Entitlement check reads only the local table

DONE WHEN

The free tier blocks a caption job at the limit, the watermark appears on every free export, and a manually-flipped entitlement row removes it.

SLICE 14 Web dashboard — writer surface

3.5 wks

The team moat depends on a surface the phone cannot provide. Different platform, same loop.

DESIGN

- Script library with folders: empty, populated, search
- Script editor with generation and version history
- Brand kit editor
- Member list, roles, invite flow
- Assignment: pick script, pick recorder
- Review inbox
- Desktop breakpoints — this is not a responsive phone layout

BUILD

- Next.js on Vercel, Supabase Auth shared session
- Roles: owner, writer, recorder
- Folders, version history
- Invite by email link
- Same Edge Functions as mobile — no duplicated logic

DONE WHEN

A writer creates ten scripts, assigns three to a recorder, and the recorder's phone receives all three.

SLICE 15 Team review loop

1.5 wks

Closes the writer-recorder circuit. The recorder path must be brutally short.

DESIGN

- Push notification → assignment inbox on mobile
- Assignment detail with a single primary action
- Submit for review
- Writer review: approve / request retake
- Comments thread
- Retake-requested state on the recorder's device

BUILD

- FCM assignment notifications
- Video upload to R2 on submit
- Review state machine: assigned → submitted → approved / retake
- Comments

DONE WHEN

Notification to recording started is two taps, and a retake request returns the recorder to the same script with the writer's comment visible.

SLICE 16 Billing

1.5 wks

Two providers writing into one entitlement table built back in slice 13.

DESIGN

- Plan picker: monthly, annual. No weekly
- Purchase in progress, success, failure
- Restore purchases
- Manage subscription entry point
- Seat management on web: add, remove, prorate notice

BUILD

- Play Billing Library 8+, solo subscriptions
- Stripe on web, team seats
- Both write to the same `subscription` table; app never queries a provider directly
- Purchase verification server-side

DONE WHEN

A Play internal-testing purchase and a Stripe web purchase both produce identical entitlement state in the app.

RISK

Play Billing only behaves correctly in builds distributed through Play. Budget time on the internal testing track — this cannot be validated from a sideloaded debug build.

SLICE 17 Hardening and device matrix

2 wks

Not a buffer. A real slice with its own acceptance criteria.

DESIGN

- Any state discovered missing during the sweep
- Onboarding — designed last, once the product is known
- Play Store listing screenshots
- Edge-to-edge inset handling audit (required by targetSdk 36)

BUILD

- Interruption matrix: call, low battery, storage full, app killed mid-export
- Offline behaviour across every network-dependent feature
- ANR and cold-start audit
- Test on both target devices, all three aspect ratios
- Crash-free rate baseline before submission

DONE WHEN

Every entry in the interruption matrix has a defined, tested outcome, and the app survives a full record-caption-clip-export cycle on the mid-range device with no ANR.

Schedule & Checkpoints

Four groups, four moments to stop and decide rather than continue on momentum.

GROUP	SLICES	WEEKS	CHECKPOINT
Foundation	00, 00.5, 01	3	Loop is working. If design-to-build took more than two rounds on slice 1, fix the handoff format before slice 2
Capture core	02, 03, 04, 05	5.5	First demo build. Real export times measured. If they are far worse than expected, revisit quality defaults now
Intelligence	06, 07, 08, 09, 10	8.5	The differentiator exists. Put it in front of five real creators before building the team layer
Business	11, 12, 13, 14, 15, 16	10.5	Monetisation and team loop complete. Play internal testing track live
Hardening	17	2	Submission-ready
Total		29.5	

The 29.5 figure exceeds the 22–27 range in the specification because it prices the design step explicitly rather than folding it into build time. Treat 22–27 as the range if design is fast and no slice needs a second round; 29.5 is the honest number with the loop run properly on every slice.

7.1 Where to cut if the schedule slips

In this order, and only in this order. Each cuts cleanly without breaking anything downstream.

- 1 Multi-take stitching (slice 05)**
 Take picker plus head/tail trim is sufficient. Saves ~0.5 week
- 2 Review loop (slice 15)**
 Keep assignment and notification; drop approve/retake and comments. Assignment alone still makes the org the billing unit. Saves ~1 week
- 3 Beauty effects (slice 08)**
 Table stakes, not a reason anyone switches. Saves ~2.5 weeks and removes the hardest performance constraint in the project

DO NOT CUT THESE

Slice 13 (watermark and metering) and slice 14 (web dashboard) look deferrable and are not. They are the two moat mechanisms. Shipping without them produces a competent teleprompter app with no defensible position — which is precisely the outcome the whole plan is arranged to avoid.

THE DISCIPLINE THAT MAKES THIS WORK

The loop is worth more than any individual slice. **Scope, prompt, design, handoff, build, sweep.** Run it identically eighteen times and the app will look and behave like one product.

The two steps that get skipped under pressure are the handoff and the sweep — the two that exist specifically to catch silent decisions you did not make. Skipping them does not save time; it moves the cost to slice 15, where it is five times more expensive to fix.